

Niklas Leinz

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Robotics PhD researcher at TUM and Volkswagen. Works on robot foundation models for contact-rich assembly (vision-language-action policies, world models) and the teleoperation and training infrastructure to run them on real industrial robots.

EXPERIENCE

Doctoral Researcher (PhD) Aug. 2025 to Present
Volkswagen AG *Wolfsburg*

- Research robot foundation models for contact-rich assembly, including vision-language-action (VLA) policies and world models; first-author paper in preparation for ICRA 2027.
- Collect multimodal demonstration datasets on UR10e robots with a custom-built VR teleoperation system.
- Run a bimanual UR robot cell and integrate an AgiBot G2 humanoid into the same software stack.
- Own the pipeline end to end: data collection, policy training, validation gates, robot deployment.
- Evaluate AI and robotics startups as technology partners for Volkswagen's global production network.

Project Engineer Oct. 2024 to May 2025
Holzbau.Tech *Rosenheim*

- Master's thesis: automated pick-and-place of timber elements with reinforcement learning and sim-to-real transfer.
- Trained DDPG+HER and PPO policies for trajectory planning with Stable-Baselines3 and Gymnasium.
- Set up a GPU-accelerated training pipeline with Docker and Jenkins.
- Supervised two working students.

Platform Engineer (Dual Study) Jan. 2022 to Sept. 2022
Bosch Security and Safety Systems *Munich*

- Built an anomaly detection system for cloud services in PyTorch and deployed it with MLflow (Bachelor's thesis).
- Added server metrics in Java (Spring Boot, Micrometer) and built Grafana dashboards for monitoring.

Application Engineer (Dual Study) Sept. 2019 to Dec. 2021
Bosch Energy and Building Solutions *Düsseldorf*

- Built a construction documentation tool used at 36 Bosch sites by over 800 employees; later ported it to Flutter.
- Published an open-source library of Bosch-styled Flutter components.

EDUCATION

Technical University of Munich Munich
Dr.-Ing. (PhD candidate) Aug. 2025 to exp. 2028

- Thesis on foundation models for robot manipulation in automotive assembly; advisor Prof. Alois Knoll.
- Teaching: Simulation-Based Machine Learning in Robotics.

Technical University of Munich Munich
M.Sc. Robotics, Cognition, Intelligence Oct. 2022 to May 2025

- Coursework: Simulation-Based Machine Learning in Robotics, LLMs in Robotics, Machine Learning, MLOps.
- Exchange semester at the Technical University of Denmark, Copenhagen (autonomous systems).

Baden-Württemberg Cooperative State University (DHBW) Friedrichshafen
B.Sc. Information Technology Oct. 2019 to Sept. 2022

SKILLS & INTERESTS

- **Robot Learning:** VLA architectures, world models, imitation learning from teleoperation, reinforcement learning (PPO, DDPG+HER), impedance and torque control, sim-to-real transfer, NVIDIA Isaac Sim/Lab, ROS 2
- **Machine Learning:** PyTorch (Lightning), Stable-Baselines3, MLflow, YOLO, OpenCV
- **Software & Infrastructure:** Python, C#, Java, Flutter, Unity, Docker, Jenkins, Linux, Terraform, AWS, GCP
- **Certifications:** KUKA Robot Programming 1 (2025), AWS Certified Cloud Practitioner (2021)
- **Languages:** German (native), English (fluent), Spanish (basic)
- **Interests:** TUM.ai Makeathon (2023); sailing, climbing, skiing, hiking